Image Processing Tutorial

Spatial Filtering & Histogram Processing

(Sheet 4 / Tutorial 4) • ECE 359

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Outline

Convolution & correlation

	1		
1	1	2	3
	0	8	6
	7	8	9

$f * H$

image — kernel (Spatial filter)

Correlation

Convolution (180° rotation)

$$H = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \rightarrow \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}$$

Correlation → use H as it is

convolution → rotation 180°

to prevent confusion H Matrices used are for directly multiply to the selected portion of the image

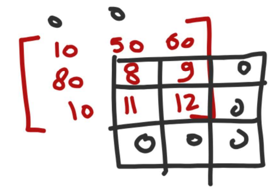
Shift filters

Image

Filter

$$\begin{bmatrix} 10 & 50 & 60 \\ 80 & 8 & 9 \\ 10 & 11 & 12 \end{bmatrix} \cdot \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{\text{Identity matrix}} = 8$$

Correlation based matrices



Shift left

$$\begin{bmatrix} 10 & 50 & 60 \\ 80 & 8 & 9 \\ 10 & 11 & 12 \end{bmatrix} \cdot \overset{H}{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}} = 9$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 10 & 50 \\ 0 & 80 & 8 \end{bmatrix} \cdot \begin{bmatrix} & 1 & \end{bmatrix} = 50$$

Shift left Convolution

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{180°} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Shift down

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

up

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Shift down $\begin{bmatrix} 10 & 50 & 60 \\ 80 & \boxed{8} & 9 \\ 10 & 11 & 12 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 50$

Shift up $\begin{bmatrix} 10 & 50 & 60 \\ 80 & 8 & 9 \\ 10 & 11 & 12 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} = 8$

1-D Filtering — Average & Median (Problem 1)

Part (a)

Consider a row of pixels with values **1, 1, 1, 1, 5, 1, 1, 1, 1**.
When we apply an average and a median filter of size 3, what would be the output values of the **5th pixel** from the left?

Part (b)

Consider a row of pixels with values **1, 1, 1, 1, 5, 5, 5, 5, 5**.
When we apply an average and a median filter of size 3, what would be the output values of the **5th pixel** from the left?

Hint

Average filter: $\hat{p} = \frac{1}{3}(p_{i-1} + p_i + p_{i+1})$ Median filter: $\hat{p} = \mathrm{median}\{p_{i-1}, p_i, p_{i+1}\}$

Solution — Problem 1

Problem 1 Solution

1, 1, 1, 1, 5, 1, 1, 1, 1

1D Averaging filter $= \frac{1}{N}\left(P_{i-1} + P_i + P_{i+1}\right)$

$$\begin{array}{|c|c|c|}\hline 1 & 1 & 1 \\ \hline \end{array} \qquad \frac{1}{3}$$

$$\uparrow$$

$$\begin{array}{cccc} & 0 & 1 & 1 \\ \frac{1}{3} \times & \boxed{1} & \boxed{1} & \boxed{1} \end{array}$$

$= (0 + 1 + 1) * \frac{1}{3} = \frac{2}{3} = 0.6 = 1$

$\rightarrow \hat{P}_5 = (1 \times 1 + 1 \times 5 + 1 \times 1)\,\frac{1}{3} = 2.3 \simeq 2$

Median

1. Sorting
2. choose middle number for centerpixel.

$$\begin{array}{|c|c|c|}\hline .. & M & ... \\ \hline \end{array}$$

$\hat{P}_5 = \text{median}\{1, 5, 1\} = \text{median}\{1, 1, 5\} = 1$

$1, 1, 1, \underbrace{1, 5, 5}, 5, 5, 5$

1. $\hat{P}_{5avg} = (1 \times 1 + 5 \times 1 + 5 \times 1)\,\frac{1}{3} = \frac{11}{3} = 2.6 \simeq 3$

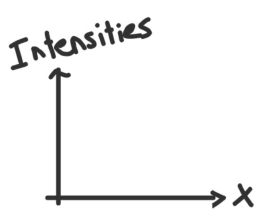
2. $\hat{P}_{5med} = 5$

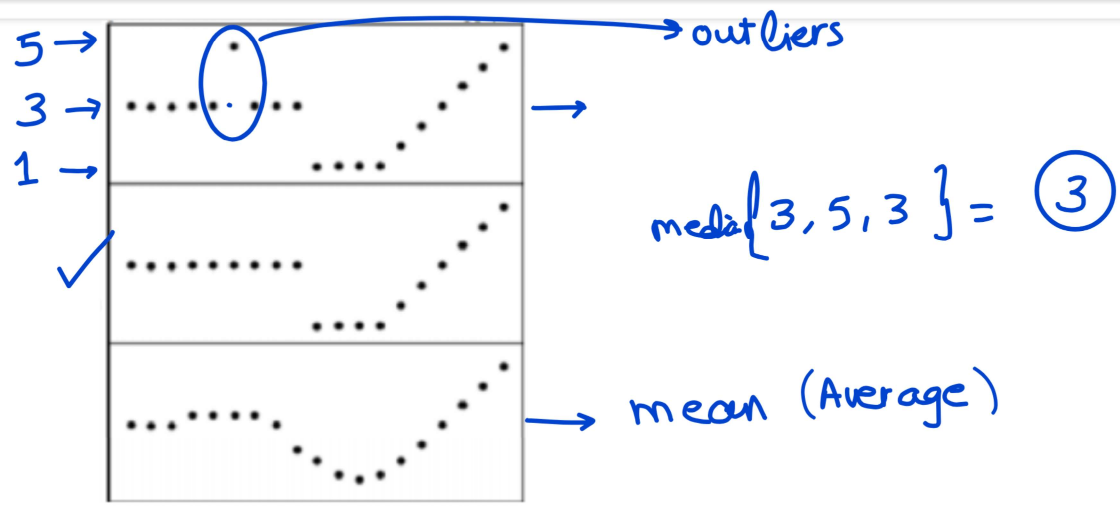
1-D Signal: Mean vs. Median Filter (Problem 2)

Problem 2

The **first** signal below is the original 1-D signal. The following two signals are the result of applying either a median or a mean filter.

State which filtered signal corresponds to each filter and give your reasoning.





Original signal (top), Filtered signal 1 (middle), Filtered signal 2 (bottom)

Reference Image (Used in Problems 3–6)

	0	1	2	3	4
0	100	102	104	102	106
1	102	105	107	106	106
2	104	**48**	108	109	110
3	105	110	104	106	109
4	107	105	103	107	105

5×5 image — indexed $[col, row]^T$ starting at $[0, 0]^T$

Indexing Convention

Coordinates are $[x, y]^T$ where $x =$ column, $y =$ row (0-based).

The value **48** at $[1, 2]^T$ is the outlier pixel.

3×3 Average & Median Filter (Problem 3)

Problem 3 — Average Filter

Using the reference image, what is the intensity at pixels $[2, 0]^T$ and $[3, 2]^T$ after a 3×3 average filter?

col row

Problem 3 (cont.) — Median Filter

What is the intensity at pixel $[2, 1]^T$ after a 3×3 median filter?

Average kernel reminder

$$H = \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Handle boundary pixels by **zero-padding** unless stated otherwise.

Solution — Problem 3

Problem 3 Solution

100	102	104	102	106
102	105	107	106	106
104	**48**	108	109	110
105	110	104	106	109
107	105	103	107	105

histogram

r_i	n
r_{109}	2
r_{110}	1
r_{106}	3

Problem 3

$$\hat{P}_{2,0} = \frac{1}{9} \cdot \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 102 & 104 & 102 \\ 105 & 107 & 106 \end{bmatrix} = 69.6 \simeq 70$$

Problem 3 (b) median at $\overset{\text{col}\ \ \text{r}}{[2, 1]}$

collect all values

Sort

choose

replace

$48, 102, 102, 104, \boxed{105}, 106, 107 \ldots$

$\hookrightarrow$ median

$\hat{P}_{21} = 105$

Average $\overset{\text{col}\ \ \text{row}}{[3, 2]}^T$

$$\hat{P}_{[3,2]} = 107.2 \simeq 107$$

Gaussian Filter (Problem 4)

Problem 4

Using the reference image, what is the intensity at pixel $[2, 2]^T$ after applying the Gaussian filter with the weighting matrix below?

$$H = \frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Note: The sum of all kernel weights equals 1
$(1+2+1+2+4+2+1+2+1 = 16)$

Solution — Problem 4

Problem 4 Solution

100	102	104	102	106
102	105	107	106	106
104	**48**	108	109	110
105	110	104	106	109
107	105	103	107	105

$$\hat{P}_5 = \frac{1}{16}\begin{bmatrix}1&2&1\\2&4&2\\1&2&1\end{bmatrix}\begin{bmatrix}105&107&106\\48&108&109\\110&104&106\end{bmatrix} = 100$$

$$= 1(105)+2(107)+1(106)+2(48)+4(108)+2(109)$$
$$+1(110)+2(104)+1(106) = 99.7 \simeq 100$$

Reverse-Engineering a Linear Filter (Problem 5)

Problem 5

After passing the reference image through an unknown linear filter, the result is shown below. Can you **identify the filter matrix** and state **how boundary pixels were handled**?

Zero padding

100	102	104	102	~~106~~
102	105	107	106	~~106~~
104	**48**	108	109	~~110~~
105	110	104	106	~~109~~
~~107~~	~~105~~	~~103~~	~~107~~	~~105~~

0	**0**	**0**	**0**	**0**
0	100	102	104	102
0	102	105	107	106
0	104	48	108	109
0	105	110	104	106

Filtered output image — Problem 5

Shift right → Shift down

$a * b$

$b * a$

Blurring + Median on Binary Image (Problem 6)

Problem 6

Image $b(x, y)$ is binary (equal numbers of 0 and 255 pixels). Two filters are applied sequentially:

- **3×3 blurring filter** with coefficients 1/5: $H = \begin{bmatrix} 0 & \frac{1}{5} & 0 \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{5} \\ 0 & \frac{1}{5} & 0 \end{bmatrix}$
- **3×3 Median filter**

(a)

If blurring is applied **before** median, what are the **possible output pixel values** of $o(x, y)$?

(b)

Give a binary image $b(x, y)$ for which $o(x, y)$ is the **same regardless of filter order**.

and viceversa

Solution — Problem 6

Problem 6 Solution

$$f(x,y) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 255 & 255 & 255 \\ 255 & 0 & 255 \end{bmatrix} \longrightarrow$$

Apply blurring filter

K: number of "255" pixels Multiplied

K	Blurred value
5	$255 \times \frac{5}{5} = 255$
4	204
3	153
2	102
1	51
0	0

possible outputs $\{0, 51, 102, 153, 204, 255\}$

(b) Any image with pixels uniformly distributed as zeros or ones.

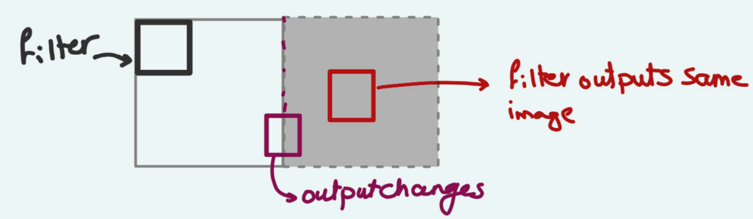
Hence, by applying blurring or median image doesn't change

→ All 0's → 3x3 neighborhood

↳ Average of all 0's is a zero

↳ same goes for median

ex: half black half white image and filter applied far from boundary.



Computing Filter Output (Problem 7)

Given: A 3×3 image region and averaging filter

Image Region

$$\begin{bmatrix} 10 & 20 & 30 \\ 40 & 50 & 60 \\ 70 & 80 & 90 \end{bmatrix}$$

Filter

$$\frac{1}{9} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Find: Output value at center pixel

$\hat{P}_5 = \frac{1}{9}[10 + 20 + \cdots + 90] =$

Median Filter Computation (Problem 8)

Given: A 3×3 neighborhood with values:

Neighborhood

$$\begin{bmatrix} 100 & 10 & 120 \\ 15 & 200 & 18 \\ 110 & 12 & 115 \end{bmatrix}$$

Tasks:

- Sort the values
- Determine the median for centre pixel
- Compare with averaging filter result

$\hat{P}_5 = \text{median}\{\cancel{10}, \cancel{12}, \cancel{15}, \cancel{18}, 100, \cancel{110}, \cancel{115}, \cancel{120}, \cancel{200}\} = 100$

$\hat{P}_5 = \frac{1}{9}[10 + 12 + 15 + 18 + 100 + 110 + 115 + 120 + 200] =$

Note: Center pixel (200) appears corrupted by impulse noise